

Sustainable Development Goals and Digital Technologies: Mapping Scientific Research

HIGHLIGHTS

- The Sustainable Development Goals (SDGs) set ambitious targets to be met by 2030. With less than 10 years left, technology is crucial to close the sustainability gap.
- This brief provides evidence on the scientific knowledge related to SDGs, Digital Technologies (DTs), and their intersection.
- China and the United States are major players in scientific research in the selected domains. Yet, EU countries together produce more scientific publications than any single country.
- Unlike other DTs, Artificial Intelligence (AI) finds applications in almost all SDGs.
- Natural Language Processing (NLP) analyses show that AI is positively associated with most SDGs, although this does not apply to SDG5 (Gender Equality), SDG10 (Reduced Inequality) and SDG16 (Peace, Justice, and Strong Institutions). The latter is the most negatively affected by digital technologies, particularly blockchain and computing infrastructure.

SUSTAINABILITY AND DIGITALIZATION

The Sustainable Development Goals (SDGs) formulated in 2015 by the United Nations (UN) General Assembly set ambitious targets to achieve a sustainable, fair, and inclusive future for people worldwide by 2030 [1]. The SDGs are an unprecedented attempt to close the sustainability gap through a focus on the main ecological, economic, and social challenges that humanity faces.

With less than 10 years left, a growing number of policy initiatives are being put in place to make sure the goals are met [2]. Science, technology, and innovation are unquestionably among the key enablers for the achievement of the 2030 Agenda's ambitions. Digitalization holds great promise for sustainable transformation.

Despite an emerging literature, the nexus between digital transformation and sustainable development remains poorly explored and understood. Knowledge on the topic is based primarily on case studies, expert opinion, and anecdotes. Growing efforts are made to map the development of digital technologies [3]. Similar initiatives have also been undertaken

to map the SDGs [4]. Yet, less explored remains the *SDGs-digitalization nexus*.

In the context of the INNOVA projects, which support the work of the Directorate-General for Research & Innovation (R&I) of the European Commission, the JRC aims to provide evidence on the scientific knowledge, measured through academic publications, related to SDGs, Digital Technologies (DTs), and their intersection. Scientific knowledge is an important driver of technological innovation, as research shows [5] strong links between knowledge creation and new technologies in various sectors such as, ICT, semiconductors, and biotechnologies.

MAPPING SCIENTIFIC RESEARCH

Data in this analysis come from the Web of Science (WoS) Core Collection, a widely used data source for science mapping. The analysis considers all scientific articles published between 2010 and 2021 in peer-reviewed journals or conference proceedings. Research on SDGs is mapped collecting publications containing either in the title, the abstract or the keywords search terms related to (at least) one of the 17 goals. Research on Digital Technologies is instead mapped using a comprehensive list of search terms inspired by recent

contributions by OECD and scholars on the mapping of advanced digital technologies [6]. The developed taxonomy considers seven macro-categories: Additive Manufacturing; Artificial Intelligence; Blockchain; Big Data; Computing Infrastructures; Internet of Things; and Robotics. Additional details on data retrieval processes are available in the Quick Guide.

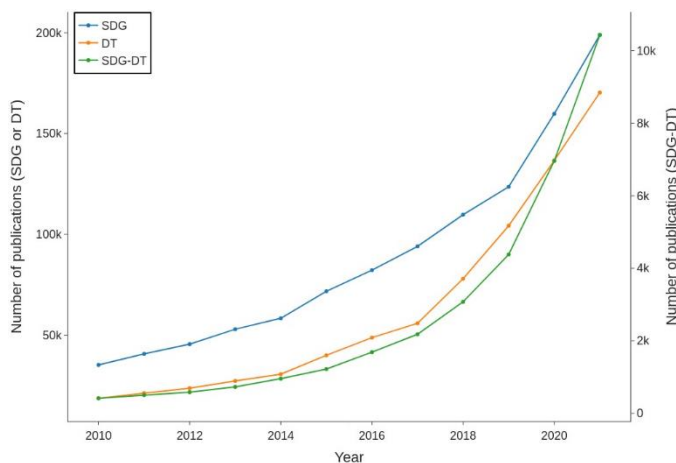
The resulting dataset covering research on SDGs comprises of almost 1.3 million publication records. Among the most frequent SDGs in the data, SDG13 (Climate action) accounts for about 22% of these publications whereas SDG3 (Good health and well-being), SDG6 (Clean water) and SDG14 (Life below water) cover about 10% of the records each. Among the less frequent, SDG1 (No poverty), SDG2 (Zero hunger), SDG8 (Decent work and economic growth), SDG9 (Industry, innovation, and infrastructure) and SDG10 (Reduced inequalities) account for just 2% each.

The resulting dataset covering research on Digital Technologies contains about 860 000 records of which more than half (~58%) are related to AI and about 110 000 to IoT (~15%) and Robotics (~17%). The least represented technology is instead Blockchain (~10 000 publications), followed by Big data and Additive Manufacturing (~50 000 publications each). Finally, scientific production covering at least one SDG and DT together account for less than 50 000 publications.

The growth of science

Scientific activity has steadily increased over the past decade. Web of Science (WoS) publications grew by 75% between 2010 and 2021. Yet, the digital and SDGs domains grew even faster during this period (Figure 1). In particular, the volume of scientific outcome related to SDGs has increased by about 450% and the volume of outcomes related to DTs by 750%.

Figure 1 – Scientific research in SDGs, DTs and their Intersection.



Source: JRC calculations on WoS data, 2010-2021. Left axis contains the number of publications related to SDGs (blue line) and DTs (orange line). Right axis details the number of publications covering both domains (green line).

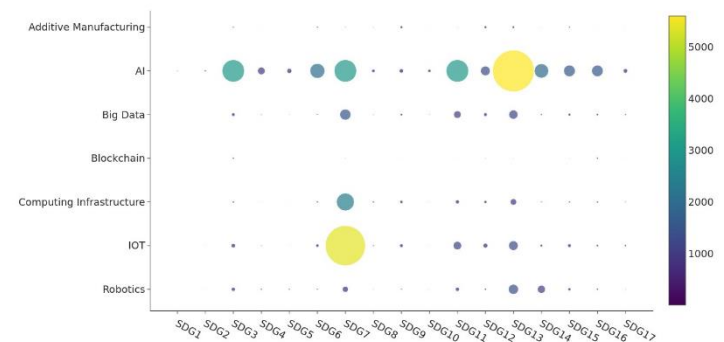
Publications jointly related to SDGs and DTs show in turn an even more pronounced growth rates (2426%), especially in recent years. This sharp increase is because scientific output at the intersection between SDGs and the recent DTs was nearly non-existent ten years ago. In 2021, by contrast, there were more than 10 000 contributions worldwide. As a result of these different dynamics, the share of publications related either to

SDGs or DTs grew from 3.6% in 2010 to 13.6% in 2021. While the volume of scientific research on SDGs remained larger than the one on DTs throughout the period, the faster growth of DTs-related scientific production implied a gradual tendency to close the gap in more recent years: the quantity of scientific research on SDGs in 2010 was 188% larger than the one on DTs; in 2021, it was 'only' 125% larger.

Goals and Technologies

Figure 2 maps the scientific knowledge related to both SDGs and DTs in the 119 combinations obtained intersecting the 17 SDGs (horizontal axis) with the seven DTs composing a digital ecosystem – i.e., a set of interconnected DTs, each differing in scope, life cycle, and degree of adoption and diffusion. First, AI plays a key role in driving the emergence of scientific research on SDGs-DTs domains, holding the largest number of publications. Unlike other digital technologies composing the digital ecosystem, it finds applications in almost all SDGs, confirming its versatility across a wide range of domains. Among SDGs, AI is particularly prominent in SDG13 (Climate change), SDG11 (Sustainable cities and communities), SDG7 (Affordable and clean energy), and SDG3 (Good health and well-being).

Figure 2 – Scientific research in the SDGs-DTs knowledge space.



Source: JRC calculations on WoS data, 2010-2021. Size and colour of the bubble is associated with the number of publications found at the intersection of each SDG and DT.

Other SDGs-DTs combinations account for fewer scientific publications, with the notable exception of the SDG7-IoT combination, which is the second largest combination after SDG13-AI. Interestingly, research in SDG7 comprises several different DTs that, beyond IoT and AI, also include scientific research on computing infrastructure and big data. This suggests that recent scientific research related to energy production, distribution and consumption is exploring new paths of knowledge re-combinations with different intertwined digital technologies. Analogous evidence, yet with lower numbers, is observed for SDG11, which beyond AI also includes sizable research on big data and IoT, and SDG13, which beyond AI also includes sizable research on big data, IoT, computing infrastructure and robotics. Finally, AI is the only technology playing a sizable role in SDG3 ('Good Health and Well-being') thanks also to the research that has been carried out in the wake of the COVID-19 pandemic (see Box 1).

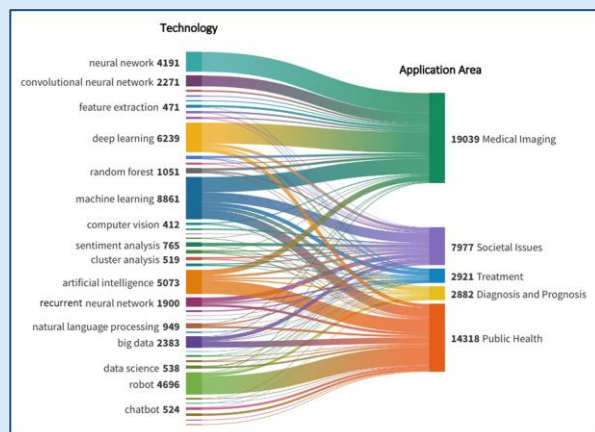
THE GEOGRAPHY OF SCIENCE

The United States (US) is the world leader in SDG-related scientific research with about 194 000 publications, followed by China with about 145 000. Notice that the EU taken as a whole, has more publications in the period 2010-2021 – i.e., about 255 000 – with a growth rate comparable to the one of China. All other countries have much lower numbers of publications, with the United Kingdom (UK) showing the largest numbers (about 61 000 publications), followed by Australia (46 000), Italy (37 000) and Germany (36 000).

Box 1: COVID-19 and scientific research

The early scientific response to the COVID-19 pandemic was a unique case of a global research effort to assess the clinical and pathogenic characteristics of the disease and formulate therapeutic strategies. In this analysis, this phenomenon is captured by the large increase observed for publications related to SDG3, which surged from about 7 000 publications in 2019 to more than 20 000 in 2020 and almost 50 000 in 2021.

Figure 3 – Occurrences of AI keywords in 5 COVID-19 domains



Source: Abbonato et al. (2022) [7].

In this context, AI has represented an effective solution to some of the challenges posed by COVID-19 (see Figure 3). Some studies confirm that AI tools have intervened at different stages of the pandemic, with applications in three main areas: molecular applications, which included, e.g., protein structure predictions, drug repurposing, and drug discovery; clinical applications, which involved, among others, autonomous and remote-controlled robots for telemedicine, image-based diagnosis, and hospital capacity planning; and social application, which involved epidemiology, where AI-based solutions have helped understand public policy interventions such as social distancing, and infodemiology, where AI has helped combat information disorders and manage information overabundance.

China has the largest number of publications on DTs over the entire period (about 189 000). In the same years, the EU ranks second with about 150 000 publications, and the US occupies the third position with about 115 000 publications. A similar number of DTs-related publications (about 10 000) is recorded yearly up to 2017 in both China and the US as well as in the

EU, while in 2021 China had more than 40 000 as compared to the EU and the US with about 30 000 and 20,000 respectively. Among other countries, South Korea and the UK show the largest number of DTs publications (respectively 33 000 and 31 000 over the entire period), followed by Germany and Italy (25 000 over the entire period).

Similarly, as in the case of SDGs, the EU as a whole leads in the production of scientific knowledge on SDGs-DTs (about 7 000 publications, roughly 21% of the total), followed by China (6 500) and the US (5 000). The trends of the EU and China are very close across years, while the gap with the US accentuated since 2016. Among other countries, the UK has the largest number of SDGs-DTs publications in the period (about 1 500), followed by Italy, South Korea, Spain, Germany and Australia (all approximately SDGs-DTs 1 000 publications).

EU regional-level strength

In absolute terms, the largest scientific production comes from Cataluña (Spain) in SDGs and Île-de France (France) in DTs, with more than 10 000 publications since 2010. Other regions with large absolute numbers (more than 6 000) of publications are: i) in SDGs, Île-de France (France), Comunidad de Madrid and Andalucía (Spain), Lombardia and Lazio (Italy), Zuid-Holland (Netherlands) and Hovedstaden (Denmark); ii) in DTs, Comunidad de Madrid (Spain), Lombardia (Italy) and Oberbayern (Germany). As for the intersection SDGs-DTs, the highest amount of scientific publications come from Lazio (Italy) with almost 600 publications, followed by Comunidad de Madrid and Cataluña (Spain) and Lombardia (Italy), with more than 200 publications.

Unsurprisingly, the weakest regions are among rural areas, some EU outermost and less populated regions such as Flevoland and Drenthe (Netherlands), Severen Tsentralen (Bulgaria), Prov. Luxembourg (Belgium), Valle d'Aosta (Italy), Burgenland (Austria) and Åland (Finland) where there are less than 100 publications on SDGs or DTs since 2010.

Considering only absolute counts may be misleading. Indeed, more populated regions are expected to have more people working in the fields of education and R&D activities, leading to larger amounts of qualified human capital in any domain. Consequently, we also report a normalized indicator, in which all figures have been normalized by total regional population to remove scale effects, at least partly.

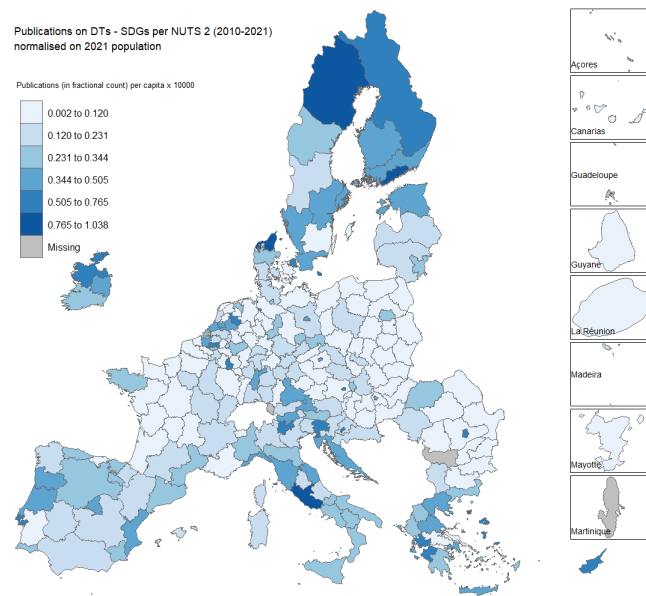
When flagging the top-15 NUTS 2 regions by the number of publications related to SDGs, DTs, and SDGs-DTs normalized by regional population, an interesting pattern emerges. Five regions from different countries are among the top-15 regions producing knowledge in all three domains. These regions are:

- Övre Norrland – SE
- Helsinki-Uusimaa – FI
- Bremen – DE
- Hovedstaden – DK
- Grad Zagreb – HR

Scientific production in the digital domain is evenly distributed across EU countries, having regions from 14 countries in the top 15. Only Denmark appears two times in the best 15 performing regions (Nordjylland and Hovedstaden). Geographical patterns are clearer when referring to SDG-related scientific production, where among the top 15 performers there are three Swedish (Övre Norrland, Östra Mellansverige, Stockholm), three Dutch (Gelderland, Utrecht, Groningen) and two Belgian regions (Prov. Vlaams-Brabant,

Région de Bruxelles-Capitale). Observing publications at the intersection of digital technologies and SDGs, four countries appear two times among the best performers: Italy (Lazio, Provincia Autonoma di Trento), Finland (Helsinki-Uusimaa, Pohjois- ja Itä-Suomi), Denmark (Nordjylland, Hovedstaden) and Greece (Voreio Aigaio, Dytiki Elláda) – see Figure 4.

Figure 4 – Scientific strength of EU regions in SDGs-DTs, 2010-2021.



Note: Number of publications computed using fractional counts (see quick guide). Numbers normalised by 2021 NUTS-2 level Population. Source: JRC calculations on WoS and Eurostat data.

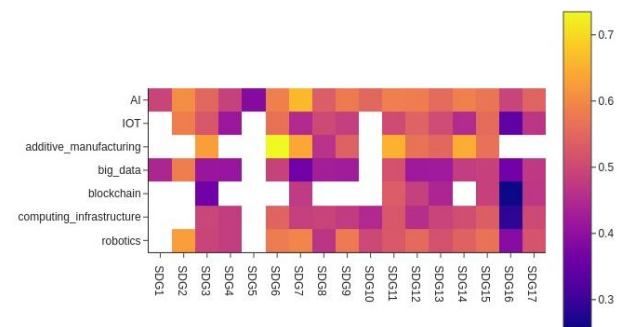
A high divergence emerges when flagging the worst-performing regions in the three domains. Six regions are in the bottom-15 list in all domains (Mazowiecki regionalny – PL, Severozápad – CZ, Vorarlberg – AT, Severen tsentralen – BG, Severozapaden – BG and Mayotte – FR). In most cases, however, being among the weakest regions in one domain is not associated to being among the weakest also in the others.

THE IMPACT OF TECHNOLOGIES ON SDGs

While it is recognized that technology has undoubtedly played a role in exacerbating some of the current challenges that humanity faces, it also has the potential to be part of the solution. A growing body of qualitative research has sought to examine the impact of digital technologies on the achievement of the Sustainable Development Goals (SDGs) [10]. In this study, we leverage NLP techniques to evaluate abstracts of scientific publications at the intersection of digital technologies and the SDGs to determine the overall contribution of digital technologies to the achievement of the SDGs. The analysis reaches three main conclusions: (i) NLP tools on the corpus of scientific publications can provide a complementary approach to qualitative research; (ii) Artificial intelligence is mostly positively associated with the SDGs, although this does not apply to SDG5 (Gender Equality), SDG10 (Reduced Inequality) and SDG16 (Peace, Justice and Strong Institutions); and (iii) SDG16 is the most negatively affected by digital technologies, particularly blockchain and computing infrastructure (see Figure 5). This analysis represents a first comprehensive step

toward understanding the complex relationship between digital transformation and sustainability.

Figure 5 – The impact of technologies on SDGs.



Source: JRC calculation on WoS data.

Conclusions

The work of the JRC in the context of the INNOVA project, which supports the work of the Directorate-General for Research & Innovation (R&I) of the European Commission, provides the first evidence of the emergence of scientific research that jointly studies SDGs and DTs.

The combination of SDGs- and DTs-related in scientific research is a recent development that has surfaced over the past ten years and is rapidly expanding. Unsurprisingly, China and the United States are major players in scientific research in the selected domains, with each country producing many publications and holding a significant share of the global research output. Yet, the European Union produces more scientific publications than any single country, including China and the United States, in the SDGs and SDGs-DTs domains and closely follows China, which leads in terms of DTs publications.

The EU's strong scientific production in areas combining knowledge on both sustainable development and digital technologies, coupled with the strong ongoing EU policy efforts to promote the development, reinforcement, and diffusion of advanced digital technologies (such as the EU Digital Strategy), is likely to foster the opportunities for the creation of EU digital solutions enabling to approach the achievement of SDGs.

Yet, fully realizing the potential of EU digital research for sustainable development requires improvements in the integration of national research systems to exploit scale and scope economies that MSs cannot achieve in isolation. This is particularly important in research on advanced digital technologies, which is a domain whose development requires solid infrastructures and specific capabilities, while being characterised by large investments worldwide. While having made some progress since the introduction of the European Research Area (ERA) in 2000, the EU scientific landscape appears fragmented [8]. The EU encourages international collaborations with other leading players in the global research community as well as collaborations with economies that are less developed. By fostering international collaborations, the EU can bring in new ideas, technologies, and talented researchers from around the world, which will help to further strengthen and diversify the ERA. In this context, Horizon Europe helps providing funding for research towards the twin transition under pillar II as well as for research infrastructures

under pillar I. Moreover, Horizon strengthens international cooperation through various actions. As collaborations are of great importance for the development of science and technology, future research could cover these aspects.

Finally, this research is a first step toward the overall objective of providing a better understanding of the complex

relationships between SDGs and DTs using the lens of science. It showed that applying NLP tools to the scientific literature on the topic might provide additional and complementary insights to qualitative analysis. It can help distinguish those Goals which are more likely to be negatively affected by DTs, allowing for future in-depth analyses.

QUICK GUIDE – Data used in the analysis come from the Web of Science (WoS) Core Collection. WoS is a widely used data source for science mapping and analysis of the “science of science”. The analysis considers all scientific articles published in peer-reviewed journals and conference proceedings. To identify scientific output in SDGs and DTs domains, the authors used a *keyword-based approach*. Queries searched for all publications over the period 2010–2021 containing at least one search term either in the title, the abstract, or among keywords. Publications on the SDGs were identified using the search queries for “Mapping Research Output to the Sustainable Development Goals (SDG)”. The project represents a comprehensive effort to map scientific output related to the 17 different SDGs. Originally developed for Scopus, the search queries are elegant constructions of keyword combinations, Boolean and proximity operators that have been adapted to WoS syntax. Digital publications were identified using a comprehensive list of search terms inspired by recent contributions on the mapping of advanced digital technologies [6]. The analysis uses *fractional counts* to analyse scientific performance of countries and European Union (EU) regions. This involved geo-localizing each publication exploiting authors’ affiliations – i.e., attributing a publication to a given country (or region) when the affiliation of at least one of its authors is in that country (or region). To avoid making assumptions about the numbers of authors per publications across different countries (regions) and scientific domains, the contribution to any given publication of each country (region) is weighted by the number of authors (n). A publication with n authors, each one located in different countries (regions), will contribute by $1/n$ to the scientific production of each country (or region). Note that multiple affiliations are counted only once when they are: (i) within the same country for the national-level analysis; and (ii) within the same region in the regional-level analysis. Finally, For NLP analysis, aspect-based sentiment analysis algorithms were used. In particular, the DeBERTa-v3-large-ABSA-v1.1 model was, a pre-trained deep learning model for sentiment analysis based on the DeBERTa-v3-large architecture – see [9] for additional details.

Related work

This policy brief is part of [the INNOVA projects](#), which support the work of the Directorate-General for Research & Innovation (R&I) of the European Commission by providing better evidence for policy-making. The INNOVA V project focuses on three interrelated research streams: R&I monitoring frameworks; innovative FDI flows and the development of green and digital technologies; and digital technologies and SDGs.

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SUGGESTED CITATION

Bianchini, S., Bottero, P., Colagrossi, M., Damioli, G., Ghisetti, C. and Michoud, K., *Sustainable Development Goals and Digital Technologies: Mapping Scientific Research*, JRC Policy Brief, 2023.

CONTACT INFORMATION

jrc-innova-project@ec.europa.eu

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